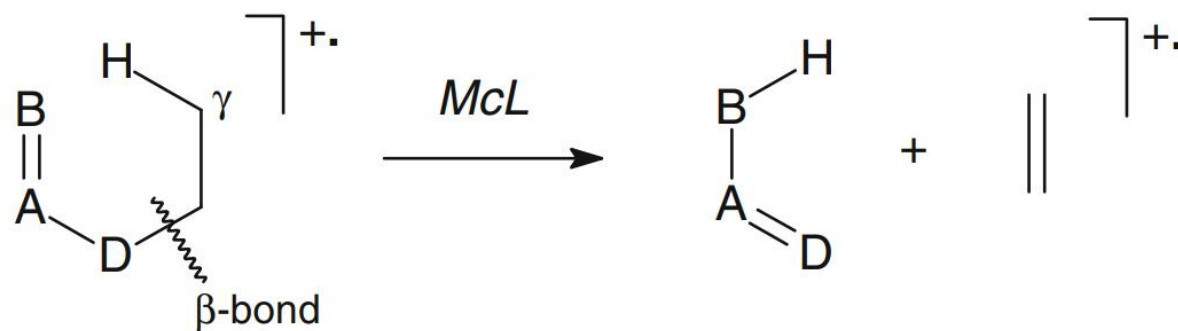


质谱解析新增知识点

McLafferty Rearrangement

- McLafferty rearrangement is one of the most important and well-known rearrangement reactions that is the γ -H shift with β -cleavage.

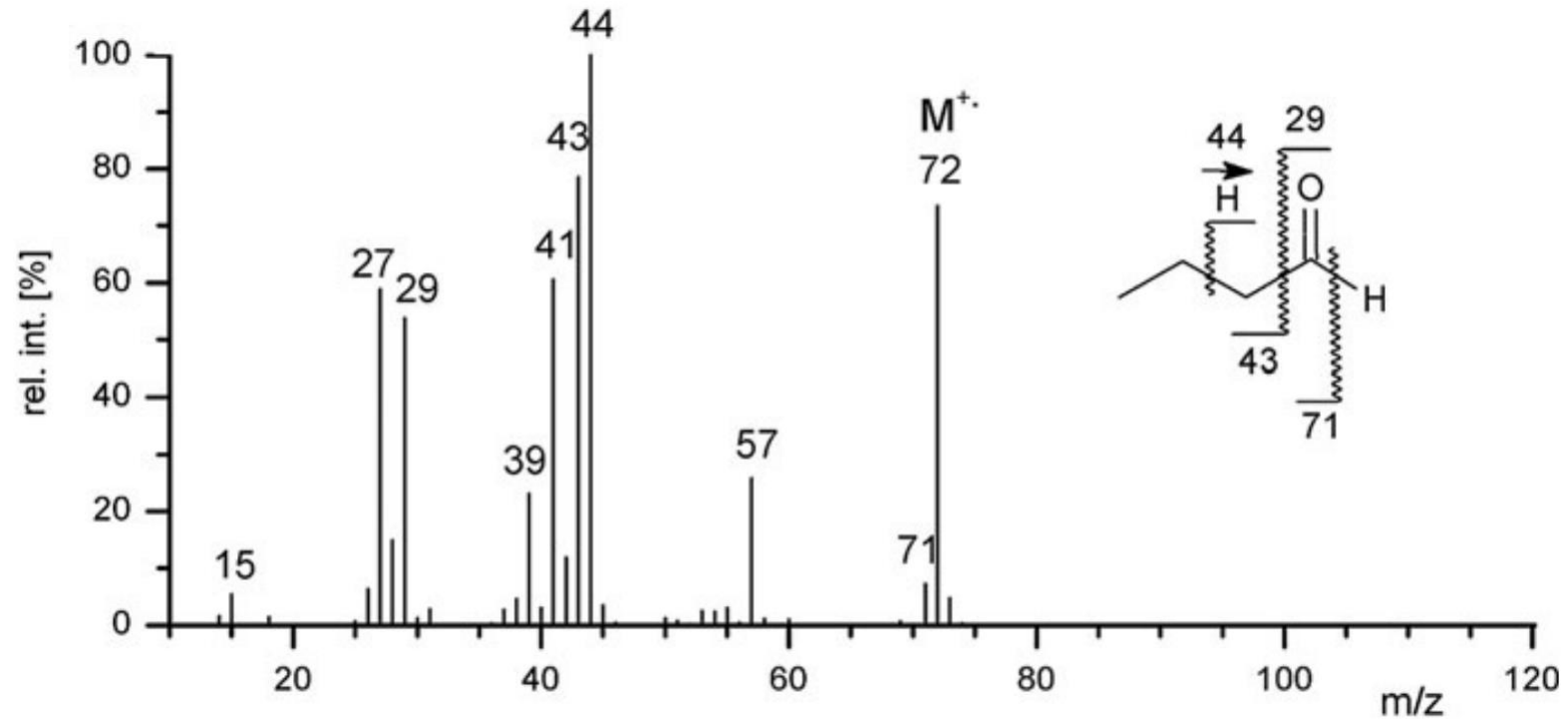
The McLafferty Rearrangement and Alkene Loss



Requirements for a McLafferty Rearrangement

1. Atoms A, B, and D can be either carbon atoms or heteroatoms.
2. Atoms A and B must be connected by a double bond.
3. There must be at least one γ -hydrogen available.
4. The γ -hydrogen is selectively transferred to atom B via a six-membered transition state.
5. The distance between the γ -hydrogen and the double-bonded atom must fall within a specific range.
6. The C γ -H bond must be in the same plane as the acceptor group.

Example: Ethene Loss from Butanal



El mass spectrum of butanal. The base peak at m/z 44 is caused by the product of the McLafferty rearrangement. Note that here the peak at m/z 57 cannot result from $[C_4H_9]^+$, but is due to $[M-CH_3]^+$

Explaining the Fragmentation

